

AI for Economic Research: Dynamic Models, Language, and Agents

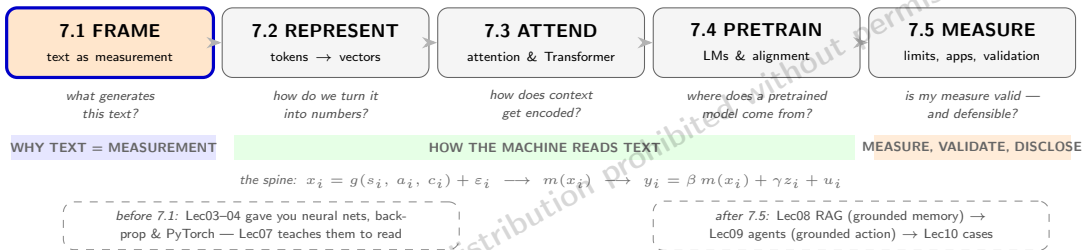
Lecture 7.1: Text as Economic Measurement

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2026-07-09 03:53:54

Lecture 7 in One Map

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The five acts, as verbs: **Frame** — pose the measurement question, $x_i = g(s_i, a_i, c_i)$; **Represent** — turn tokens into numeric vectors; **Attend** — let surrounding context reshape each token's meaning; **Pretrain** — learn that structure from raw text at scale, then align it; **Measure** — turn the machinery into a validated, defensible economic variable. *Two case threads run through all five acts —*

Thread A: the EPU index (*Baker–Bloom–Davis 2016*), the transparent dictionary baseline every method must beat;

Thread B: FOMC hawk–dove (*central-bank communication*), the running example for embeddings, attention, and validation. Watch for them.

Text as Data

Why language became an economic measurement problem

The Opening Hook: Reading Uncertainty out of Newspapers

In 2016, three economists asked: can you read policy uncertainty out of the newspaper?

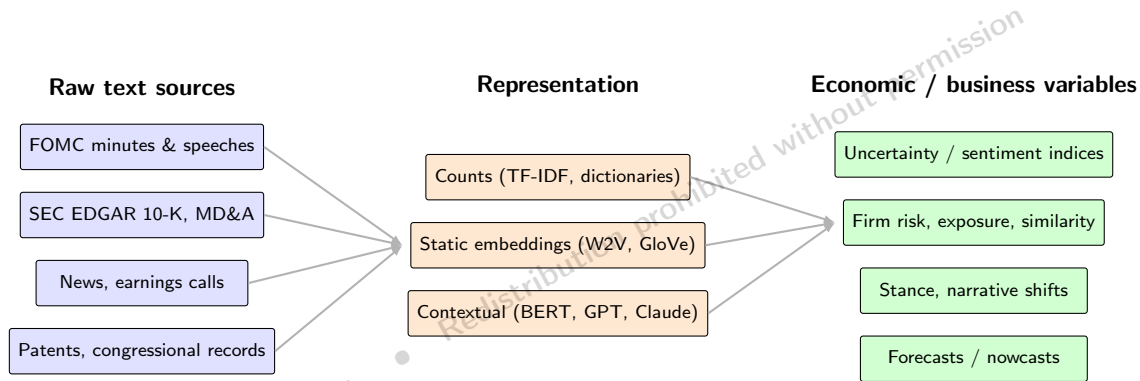
- Baker, Bloom & Davis (2016, *QJE*): scan 10 major US newspapers, month by month, and count articles containing **all three** of:
 - {economic or economy}, AND
 - {uncertain or uncertainty}, AND
 - one of {Congress, deficit, Federal Reserve, legislation, regulation, White House}
- Scale by each paper's total output, standardize, average across papers, index to 100:

$$\text{EPU}_\tau = 100 \times \frac{u_\tau}{\bar{u}}, \quad u_\tau = \frac{1}{10} \sum_{p=1}^{10} \frac{c_{p\tau}/N_{p\tau}}{\sigma_p}.$$

- No statistical agency publishes “policy uncertainty” — it is not directly observed anywhere. Yet in a VAR, EPU spikes precede investment and employment declines.

The move that matters: unstructured language became a *regression-ready variable*. This act asks what that move takes — and where it can go wrong. (*Thread A begins.*)

Text as Economic Data — The Big Picture



The lecture builds the middle column rung by rung: Act 2 counts → static embeddings, Act 3 contextual machinery, Act 4 how those models are trained — and Act 5 turns the machinery into the right-hand column, with validation.

Why Economists Care About LLMs

- **Text as data** — modern economic research increasingly leverages textual sources:
 - Corporate filings (10-K, 10-Q, MD&A)
 - Central bank speeches, FOMC minutes, ECB statements
 - Earnings calls, analyst reports, regulatory filings
 - News, social media, congressional records
- **Applications:**
 - Sentiment analysis for financial forecasting
 - Policy stance detection and uncertainty indices
 - New text-derived macroeconomic indicators
 - Summarization of long, technical documents
- **Why LLMs in particular:**
 - Capture nuanced language and domain-specific jargon
 - Handle long-range context (sentences, paragraphs, whole documents)
 - Provide a unified architecture for many tasks

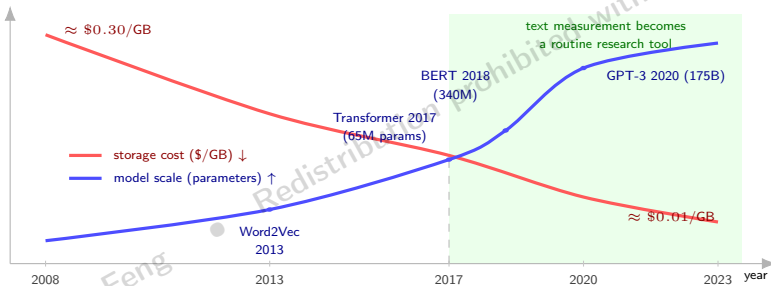
Three Roles of Text in Economic Research

Role	What text is	Example
Measurement tool	A way to construct previously unobserved variables	Baker–Bloom–Davis (2016) Economic Policy Uncertainty index from newspapers
Behavioral outcome	A response variable revealing decisions or institutional change	Hansen, McMahon & Prat (2018): how FOMC transparency reform changed committee speech
Treatment variable	A quasi-experimental input affecting behavior	Central bank communication shocks and inflation expectations

Take-away: the same text can play very different econometric roles. Choose your representation method to match the role—transparency matters for measurement, while subtle context matters for behavior modeling.

Why Now? The Feasibility Window

Two curves crossed around 2017: the cost of handling text collapsed while model capability exploded.



Both quantities on log scales; curves are stylized interpolations through the marked real data points. Storage costs after McCallum's disk-price series; model sizes from Vaswani et al. (2017), Devlin et al. (2018), Brown et al. (2020).

The Spine

A data-generating process for text — and the two hazards it creates

An Identification-First View of Text

- **Engineering perspective first:** to a machine, a document is just a character string. But every string was *produced* — by a person, in a situation, for an audience. Before any econometrics, ask: what generated this string?

- **The data-generating process (DGP):**

$$x_i = g(s_i, a_i, c_i) + \varepsilon_i$$

- s_i : the latent **economic state** we want to measure (policy stance, sentiment, firm risk)
- a_i : the speaker's **strategic framing** — audience management, spin, emphasis, tone chosen for effect
- c_i : **institutional context** — venue, legal disclosure requirements, format conventions
- **Why this matters for a downstream regression** $y_i = \beta m(x_i) + \gamma z_i + u_i$:
 - If framing a_i is correlated with the error u_i , the text feature $m(x_i)$ is *endogenous* — the same omitted-variable problem as always, but hidden inside the language
 - Measurement error is non-classical: it correlates with the true state s_i , biasing $\hat{\beta}$
- **Two research uses:** (a) **Measuring** s_i — choose representations that filter out a_i ; (b) **Studying strategic communication** — a_i is your outcome variable (e.g. Hansen, McMahon & Prat 2018).

Two Firms, Same Risk, Different Words

The endogeneity hazard, made concrete. Imagine two firms facing the *same* objective political risk s_i .

- Firm A's management discloses risks proactively — building a reputation for transparency.
- Firm B's management avoids politically sensitive topics — afraid of spooking the market.
- Same s_i , different strategic framing $a_i \Rightarrow$ different texts $x_i \Rightarrow$ different measured risk $m(x_i)$.

- If the disclosure choice a_i correlates with anything in u_i (governance quality, investor base, financial health), $m(x_i)$ is **endogenous** — before any fancy NLP even enters.
- And the error is **non-classical**: correlated with the true state s_i , so the usual “attenuation toward zero” intuition fails — the bias direction is unpredictable.
- **Discussion**: what institutional features or external variables could strip out the strategic-disclosure component? (One answer — an observable behavior like lobbying spending — appears two frames ahead.)

Three Classic Cases

Dictionary → topic model → bigrams — and what each teaches about validity

Case 1 — EPU: The Validation Standard (Thread A)

Why is a word-count index taken seriously? Because every step of it was audited.

- **Human audit:** a trained team hand-coded 12,009 articles (1900–2012) under a 65-page guide; the six policy terms were selected from $\sim 32,000$ candidate combinations to minimize false positives + false negatives.
- **Machine vs. human:** correlation 0.86 (quarterly, 1985–2012) and 0.93 (annual, 1900–2010); deviations uncorrelated with GDP growth.
- **Robustness:** dropping any one policy term barely moves the series; a 1,500-newspaper daily version correlates 0.85 with the 10-paper index.
- **And still — the identification critique:** does EPU *cause* downturns, or do downturns make journalists write “uncertainty”? The VAR ordering (EPU first) is a maintained assumption, not an identification argument.

Dictionary methods: transparent, replicable, auditable — but context-blind, and correlation with the cycle is built in. The “count + standardize + audit” recipe is the benchmark every fancier method in this lecture must beat.

Case 2 — FOMC Transparency: Language as an Outcome (Thread B)

Sometimes the language itself is the dependent variable.

- October 1993: the Fed announces that FOMC meeting transcripts will be published (with a lag). Before: deliberation off the record. After: every word eventually public.
- Hansen, McMahon & Prat (2018, QJE): extract topic mixtures from the transcripts with LDA (a probabilistic topic model), then run a **difference-in-differences** around the reform — inexperienced regional presidents vs. experienced members.
- **Two effects of being watched:**
 - *Discipline*: members prepare more — discussion more focused, more quantitative;
 - *Conformity*: expressed views converge toward the chair — opinion diversity falls, and information aggregation may suffer.
- **Where the identification lives:** in the institutional reform, not in the text method. LDA only *operationalizes* “deliberation” into measurable statistics — here the strategic language a_i is the outcome of interest.

In the DGP's terms: research use (b) — studying strategic communication itself. Same equation, different target.

Case 3 — Firm-Level Political Risk: Micro Heterogeneity

Macro indices hide where the action is.

- Hassan, Hollander, van Lent & Tahoun (2019, QJE): count **bigrams** (two-word phrases) drawn from a political-text pattern library in quarterly earnings-call transcripts, weighted by proximity to “risk” words $\Rightarrow$ firm-quarter index PRisk_{it} .
- **Headline finding:** over 90% of the cross-sectional variance in political risk is *firm-level* — not industry, not economy-wide. Relying on macro EPU alone systematically understates political risk’s effect on investment.
- One s.d. higher PRisk associates with: investment rate -0.5pp , employment growth -0.4pp , lobbying probability $+26\text{pp}$.
- **External validity anchor:** lobbying spending — an observable, independent behavior — correlates strongly with the text measure. (The answer to the previous section’s discussion question.)
- Fixed-effects regressions deliver associations, not causal effects: PRisk may co-move with unobserved firm characteristics.

Method lesson: bigrams keep phrase-level meaning (“supply chain”, “trade war”) that single-word counts destroy — the first step up the representation ladder

The Same Program on Chinese Text

Every ingredient has a Chinese counterpart — plus two validity twists of its own.

- **China EPU** (Davis, Liu & Sheng 2019): the Baker–Bloom–Davis recipe on *People's Daily* and *Guangming Daily*, monthly since 1949.
 - Twist 1 — state-media sourcing: reporting intensity is managed in politically sensitive periods, so the index may be *understated exactly when uncertainty peaks*.
 - Twist 2 — vocabulary drift: policy terms are era-specific (“supply-side reform”), and keyword translation equivalence cannot be validated against the English index.
- **Data infrastructure map:** CNRDS / CSMAR / Wind $\approx$ EDGAR / Compustat / CRSP — A-share annual-report MD&A sections, investor-Q&A transcripts, announcements, financial news, in bulk.
- **Disclosure incentives differ by ownership:** SOE executives write under tighter political-compliance constraints than private-firm executives, so “innovation” or “reform” counts may track policy signaling rather than organizational reality. Validate text measures *separately* by ownership type.

Sources: Davis, Liu & Sheng (2019); CNRDS / CSMAR platform documentation.

Preview: The Machine Ahead

One equation will power everything in Acts 2–4

A Brief History: RNN → Transformer → GPT

- **2003:** Bengio et al. neural language model — continuous word vectors
- **2013:** Mikolov et al. *Word2Vec* — dense word embeddings at scale
- **2014:** Pennington et al. *GloVe* — global co-occurrence matrices
- **2017:** Vaswani et al. “*Attention Is All You Need*” — the Transformer replaces RNNs/LSTMs
- **2018:** Devlin et al. *BERT* (encoder); Radford et al. *GPT* (decoder)
- **2020–2022:** GPT-3, instruction tuning, RLHF → ChatGPT moment
- **2023–2026:** GPT-4/5, Claude (3–4), Gemini, LLaMA-3, DeepSeek; long-context (100k–1M tokens), open-weights ecosystem, agentic systems

Driver of the shift: attention scales better than recurrence, both in compute (parallel) and in quality (long-range dependencies).

The Autoregressive Principle

- An LLM is fundamentally a **next-token predictor**:

$$p(t^{(n+1)} \mid t^{(1)}, t^{(2)}, \dots, t^{(n)}).$$

- **Generation:** sample $t^{(n+1)}$ from this distribution, append to the context, and repeat.
- **What this single equation enables** (intuition on the next two slides; the full revisit comes at the end of Act 4, once pretraining and fine-tuning are on the table):
 - **Summarizing 10-K filings or FOMC minutes** — prompt with document + instruction; the model predicts summary tokens one by one
 - **Generating narrative forecasts** conditional on macro state — condition on data in the prompt; next tokens extend the narrative
 - **Producing scenario analyses in plain language** — the scenario is the context; prediction fills in the implications
 - **Writing and debugging code** for empirical work — same objective over a code corpus yields code completions
- **Key insight:** “everything is next-token prediction” is mathematically tiny but pedagogically and practically enormous.

Intuition: Summarizing a 10-K Filing

- **Task:** condense 50 pages of an SEC 10-K risk section into 3 readable bullets.
- **How the autoregressive equation achieves this:**
 1. Concatenate the instruction and document into the prompt:
■ Summarize the following 10-K risk section in 3 bullets:\n\n[text]\n\nSummary:■
 2. The model predicts token $t^{(n+1)}$ from the full prompt; this becomes the first word of the summary
 3. Each subsequent token is predicted given everything before it (including the already-generated summary words)
 4. Generation stops at a period or end-of-sequence marker
- **Why does it work?** During pretraining the model read millions of (long document, short summary) pairs. The next-token distribution has implicitly learned: *“after an instruction like this, produce a concise factual summary.”*
- **Economic punchline:** a 50-page MD&A section is compressed into 3 bullets in under one second — and the same pipeline scales across thousands of filings.

Intuition: Extracting Stance from FOMC Minutes

- **Task:** classify each sentence in FOMC minutes as *hawkish*, *dovish*, or *neutral*.
- **How the autoregressive equation achieves this:**
 1. Prompt: ■Sentence: [sentence]\nStance (hawkish/dovish/neutral):■
 2. The model predicts the next token; because the vocabulary contains these exact words, the top-probability token *is* the label
 3. With fine-tuning on labeled FOMC data, the next-token distribution concentrates on monetary-policy-relevant labels
- **What fine-tuning changes:** the base model already understands “hawkish” and “dovish”. Fine-tuning shifts the distribution so that, after a central-bank sentence, stance-label tokens are consistently most probable.
- **Economic application:** automated stance extraction across decades of FOMC minutes — enabling the kind of large-scale communication analysis in Hansen, McMahon & Prat (2018) without manual coding. (*Thread B, to be continued in every act.*)

Bridge: From Frames to Representations

This act fixed the question; the next three acts build the measurement machine.

- From here on, every corpus is read as draws from $x_i = g(s_i, a_i, c_i) + \varepsilon_i$ — and every text variable as an $m(x_i)$ whose two hazards (endogenous framing, non-classical error) must be argued away, not assumed away.
- The EPU baseline works by *counting*. But counts treat every word as its own orthogonal dimension: to a bag of words, *unemployment* and *joblessness* are unrelated strings — and so are a Chinese corpus's “yanghang” (central bank) and “Lianchu” (the Fed).
- **Act 2 asks:** how do we turn tokens into numbers that know *central bank* $\approx$ *Federal Reserve*? First counts done right (T2a), then dense vectors (T2b).

Arc: **T1 FRAME** $\rightarrow$ *T2 REPRESENT* $\rightarrow$ *T3 ATTEND* $\rightarrow$ *T4 PRETRAIN* $\rightarrow$ *T5 MEASURE*.